DAY 22 | PRE-READ

Searching by Meaning

A light read before Day 22. About 15 minutes, no code required.

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Why read this?

On Day 18 you saw that an AI assistant needs the right passages from your documents. Today you learn how computers find passages by meaning, not just by matching words, how to make that search fast and accurate, and when a different approach is cheaper.

1. Meaning as a location

Imagine a giant map where every sentence has a location, and sentences with similar meanings are placed close together. "How many holidays do I get?" sits right next to "Employees receive 18 days of paid leave", even though they share almost no words. The location of a sentence on this map is its **embedding**.

2. A library for these locations

A **vector database** stores millions of these locations and finds the nearest ones to a question in milliseconds. Like a librarian who knows the shelves well, it does not check every book; it goes straight to the right area. This shortcut is called **approximate nearest neighbour** search.

3. Two kinds of search, better together

Combining both is called **hybrid search**, and it is what most serious systems use.

Search type	Strength	Weakness
Keyword search	Exact terms like error code VPN-417 or a plan name	Misses different wording for the same idea
Meaning search	Understands paraphrases and synonyms	Can miss rare exact codes

4. A second opinion

Search returns a shortlist. A **re-ranker** then reads each candidate carefully against the question and reorders them, like a hiring manager reviewing the shortlist a recruiter sent.

Key idea: good answers come from good evidence. Most RAG problems are search problems.

5. Four ways to teach a model NEW (v4)

You can hand the model the right pages (**RAG**), retrain it on examples (**fine-tuning**), paste everything into one long message (**long context**), or let the provider remember a long opening you send every time (**caching**). Each has a different cost. Tomorrow you learn how to choose by size,

freshness and token cost.

6. Think about these before class

- Think of a time a search engine misunderstood you. Was it a keyword or a meaning problem?
- Which exact codes or plan names in telecom would keyword search handle better?
- If an FAQ changes every week, would you retrain a model or update a document? Why?

Imi (意味): meaning